

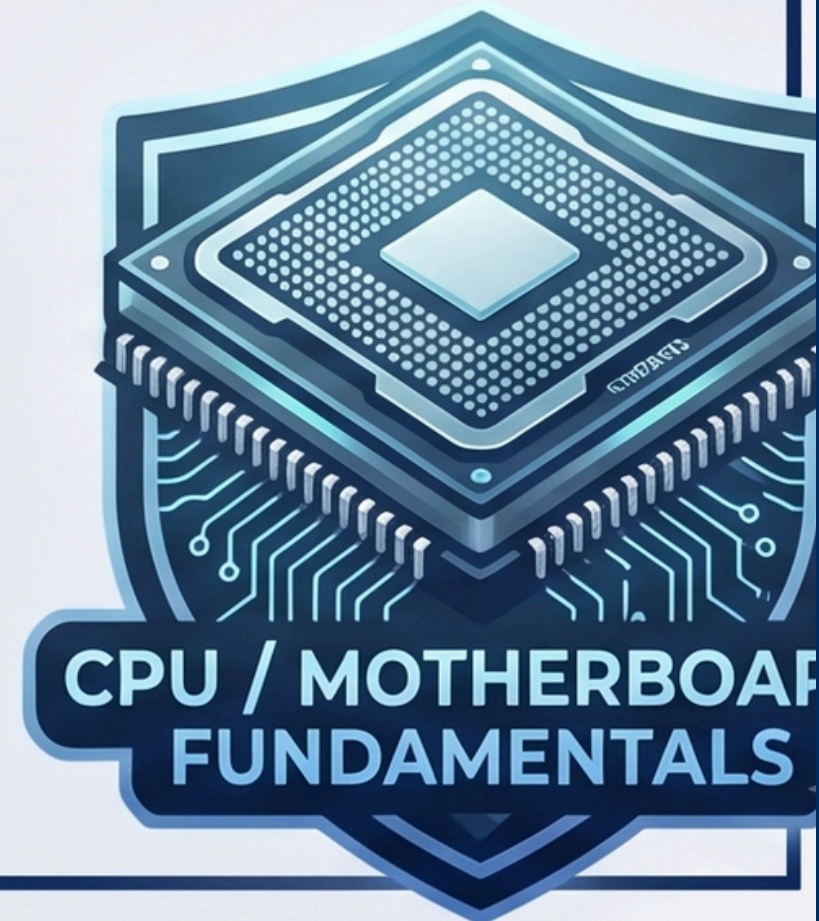


NAME:-HET ACHARYA

Course: Diploma in Cyber Security

DAY 2 – MOTHERBOARD AND CPU

Topic: Motherboard and CPU



WHAT IS A MOTHERBOARD?

The Motherboard: The PC's Main Connection Board

A motherboard is the main circuit board inside a computer that connects the important hardware components together.

It provides places for the CPU, RAM, GPU, storage devices and other hardware to connect. It also carries data and power between different parts of the computer.

Think of it like this:

Imagine a city.

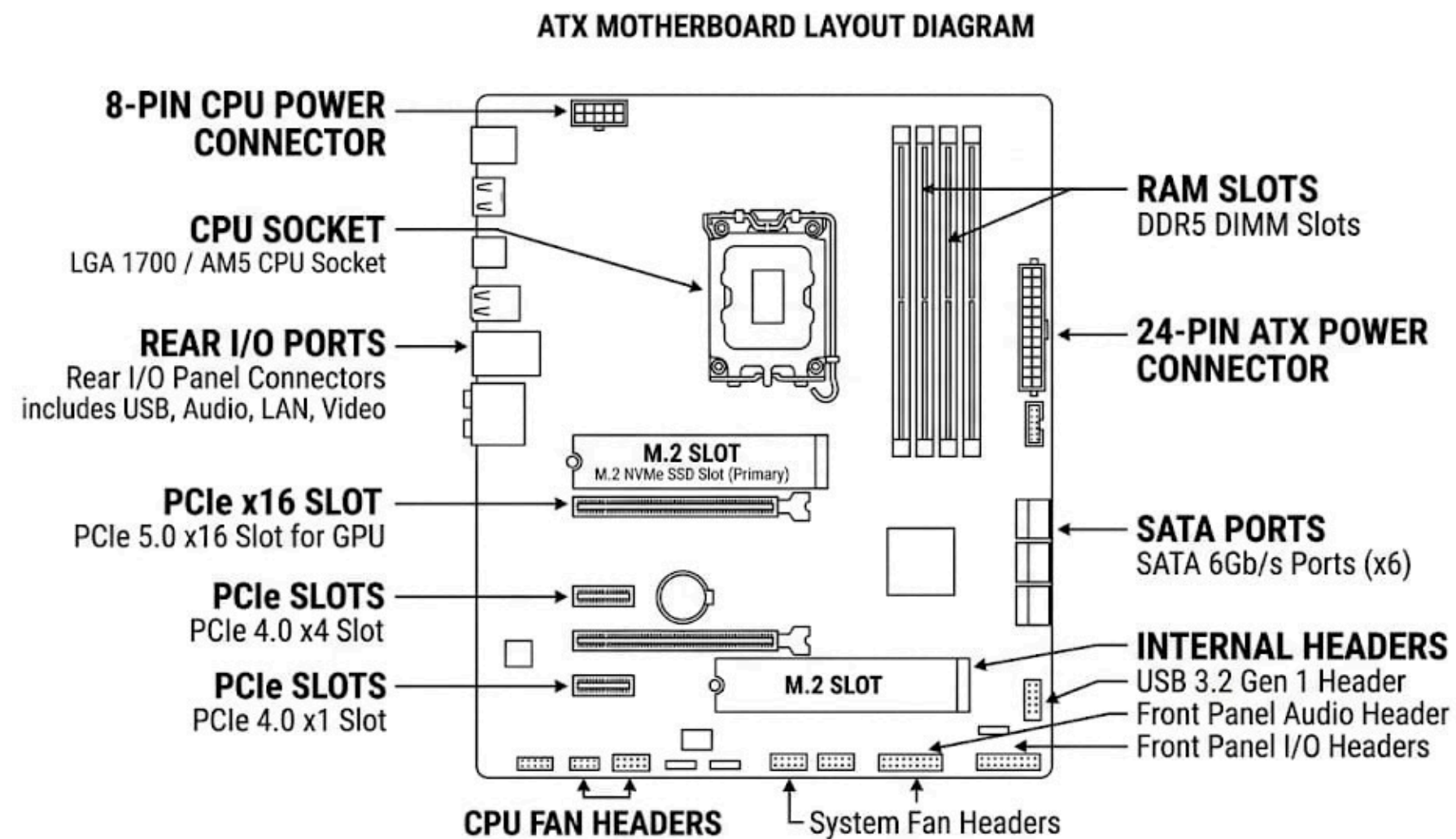
- **CPU = Main worker**
- **RAM = Temporary workspace**
- **GPU = Graphics department**
- **Storage = Data warehouse**
- **Motherboard = Road and connection system**

Without the motherboard, these components would not be able to work together as one computer.

ATX MOTHERBOARD DIAGRAM

Label at least 8 parts

use an ATX motherboard image and add arrows to these parts:



The 9 parts to label:

1. CPU Socket
2. RAM Slots
3. 24-Pin Power Connector
4. PCIe Slot
5. Second PCIe Slot
6. M.2 Slot
7. SATA Ports
8. USB Headers
9. CPU Power Connector

You only need 8, but labeling 9 makes the assignment look more complete.

WHAT DO THESE SLOTS AND PORTS DO?

1. CPU Socket

The CPU socket is the location where the processor is installed. The socket type must match the CPU, otherwise the processor cannot be installed correctly.

Easy memory: CPU goes here.

2. RAM Slots

RAM slots hold the computer's memory modules. Desktop motherboards commonly have multiple RAM slots so that additional memory can be installed.

Easy memory: RAM goes here.

3. PCIe Slots

PCIe (PCI Express) slots are expansion slots used for devices such as graphics cards and other expansion cards.

A large PCIe x16 slot is commonly used for a dedicated GPU.

Easy memory: Add extra hardware here.

4. SATA Ports

SATA ports are used to connect compatible storage devices such as SATA HDDs and SATA SSDs to the motherboard.

Easy memory: Storage connection.

5. M.2 Slot

An M.2 slot can support compact devices such as NVMe SSDs, depending on the motherboard's specifications.

Easy memory: Small, fast storage connection.

6. USB Headers

USB headers are internal motherboard connectors used to connect USB ports on the front or top of a PC case.

Easy memory: Connects case USB ports.

7. 24-Pin Power Connector

This connector supplies major power from the power supply unit (PSU) to the motherboard.

Easy memory: Main motherboard power.

8. CPU Power Connector

This connector provides additional power from the PSU to the CPU area.

Easy memory: CPU power.

CPU SOCKET TYPES

What is a CPU Socket?

A CPU socket is the physical interface on a motherboard where a processor is installed. The important point is that every CPU does not fit every motherboard. The processor and motherboard must support the same socket and platform requirements.

Three examples:

CPU Socket	Example	Compatible CPU
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LGA1151		
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	Intel Core i7-9700*	
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AM4		
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	AMD Ryzen 5 3600	
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LGA1200		
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	Intel Core i5-10400	
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Important note

*LGA1151 covers different Intel CPU generations, and compatibility depends on the specific motherboard chipset/BIOS. So, simply seeing “LGA1151” is not always enough to confirm compatibility.

Easy way to remember:

Socket = CPU's physical home

The CPU needs the correct “home” before it can work with the motherboard.

CPU VS CHIPSET

CPU – The Main Processor

The CPU performs instructions and calculations needed by software and the operating system.

It handles tasks such as:

- Running applications
- Processing instructions
- Performing calculations
- Managing many system operations

Simple idea:

CPU = Does the processing

Chipset – The Communication Manager

The chipset is a group of motherboard controller functions that helps manage communication between the processor and various motherboard-connected devices.

Depending on the platform, the chipset can provide or manage features related to:

- USB
- SATA
- PCIe connectivity
- Additional expansion features
- Other motherboard I/O

Simple idea:

Chipset = Helps manage connections and features

Easy comparison

CPU

Chipset

Processes instructions

Helps manage platform connectivity

Performs calculations

Provides/controls motherboard features

Runs software tasks

Helps devices communicate with the system

WHAT HAPPENS WHEN I OPEN SPOTIFY?

Example: Opening Spotify

Let's imagine I click the Spotify icon on my computer.

Step 1 – CPU

The CPU receives and processes the instructions required to launch the application.

Step 2 – RAM

The application and its currently needed data are loaded into RAM so they can be accessed quickly while Spotify is running.

Step 3 – Chipset and Motherboard Connections

The motherboard platform provides the connections that allow the CPU and other hardware to communicate with devices and system components.

Step 4 – Storage

Spotify's application files are stored permanently on an SSD or HDD. The required files are read from storage when the application starts.

Step 5 – Output

The processed information is sent to the appropriate hardware, such as the audio system, so we can hear the music.

Simple flow:

Click Spotify



CPU processes instructions



Data is loaded into RAM



System components communicate through motherboard/platform connections



Audio output plays music

REAL MOTHERBOARD IDENTIFICATION

Identify these parts on a real motherboard photo

For this page, add a clear real motherboard photo in Canva and use circles/arrows to identify at least five parts.

Label these:

① CPU Socket

The processor is installed here.

② RAM Slots

Memory modules are installed here.

③ PCIe Slot

Used for expansion cards, especially a dedicated graphics card.

④ SATA Ports

Used to connect SATA storage drives.

⑤ M.2 Slot

Used for compatible M.2 devices such as NVMe SSDs.

Optional extra labels:

⑥ 24-Pin Power Connector

Main motherboard power connection.

⑦ CPU Power Connector

Provides power for the CPU area.

⑧ Rear I/O Ports

External ports such as USB, LAN and audio.

QUICK REVISION

Remember the Motherboard in 5 Lines

CPU Socket → Where the processor sits.

RAM Slots → Where system memory is installed.

PCIe Slots → Where expansion cards can be installed.

SATA/M.2 → Connections for compatible storage devices.

Power Connectors → Bring power from the PSU to the motherboard and CPU.

My One-Line Definition

“A motherboard is the computer's main connection platform, bringing different hardware components together so they can receive power, exchange data and work as one system.”

CONCLUSION

The motherboard is not the component that performs every task itself. Its main importance is that it provides the platform and connections that allow the CPU, RAM, storage, GPU and other devices to work together.

The CPU handles processing, while the chipset and motherboard platform provide important connectivity and I/O features. Understanding these parts makes it easier to build, upgrade and troubleshoot a computer.